

# FAISAL ASLAM

Senior AI/ML Engineer | Generative AI, Agentic Systems & RAG | AI Solution Architect

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## PROFESSIONAL SUMMARY

Senior AI/ML Engineer and Solution Architect with **5+ years** building and shipping production Generative AI systems across Large Language Model (LLM) applications and Computer Vision. Specialized in agentic AI, multi-agent orchestration (LangGraph, CrewAI), Retrieval-Augmented Generation (RAG), and autonomous workflows deployed on AWS, GCP, and Azure. Currently Senior AI Engineer and Team Lead at Linvex Solutions, owning end-to-end architecture, cloud infrastructure, and cross-functional delivery. Proven record of turning complex requirements into reliable, compliant (HIPAA, GDPR) AI products with measurable business impact.

## CORE COMPETENCIES & TECHNICAL SKILLS

**Generative AI & LLMs:** Large Language Models (LLMs), Prompt Engineering & Optimization, Fine-Tuning (LoRA), GPT-4 / GPT-4o, Claude, Gemini, LLaMA 3.2, Pixtral, Multimodal AI, Multi-LLM Orchestration

**Agentic AI & Orchestration:** AI Agents, Agent Engineering, AI Automation, AI Workflow Orchestration, Multi-Agent Systems, LangGraph, CrewAI, LangChain, n8n, ReAct, Tool Calling / Function Calling, Structured Outputs, Model Context Protocol (MCP), Guardrails, Self-Correcting Agents

**RAG & Retrieval:** Retrieval-Augmented Generation (RAG), Agentic RAG, Hybrid RAG, GraphRAG, LightRAG, Vector Search (FAISS), Neo4j Knowledge Graphs, Azure Document Intelligence, Natural Language to SQL

**Evaluation & Observability:** LangSmith, Evals / Evaluation Frameworks, LLM Evaluation, AI Observability, Structured Logging

**Computer Vision:** YOLOv8, DeepSORT, Face Recognition, Object Detection & Tracking, Image Segmentation, License Plate OCR, RTSP / Video Streams

**Voice & Speech AI:** OpenAI Whisper (ASR), Speaker Diarization, Google STT / TTS, SIP / VoIP Integration

**Backend & APIs:** Python, FastAPI, Flask, Django, REST APIs, WebSockets, Async, Salesforce API, Google Maps API, SQL

**Cloud & MLOps:** AWS (SageMaker, Lambda, EC2, S3, IAM, ECS/Fargate), GCP (Vertex AI, Cloud Run, Pub/Sub), Azure (Azure OpenAI, Azure AI Foundry), Docker, CI/CD, GenAIOps

**Web3 & Blockchain:** Conversational Trading Agents, Solana / Base / Worldchain, DeFi, Non-Custodial Wallet Integration, On-Chain Tool Execution, DexScreener / Hyperliquid APIs, Server-Sent Events (SSE)

**Compliance & Governance:** HIPAA (PHI Security, RBAC, Audit Logging), UK GDPR (Data Minimization, Consent Management, Right-to-Erasure)

## PROFESSIONAL EXPERIENCE

**Senior AI Engineer (Team Lead)** — Linvex Solutions

Mar 2025 – Present

Lahore, Pakistan · Full-time · On-site

**Vyera AI — Competitive Intelligence Multi-Agent Platform** LangGraph · Multi-Agent · Firecrawl · FastAPI · AWS · UK GDPR · Live: [vyera.ai](https://vyera.ai)

- Architected an AI-powered competitive intelligence platform for the UK market that autonomously discovers competitors, scrapes their data via Firecrawl, and synthesizes strategic insights through a multi-agent LangGraph supervisor graph.
- Built specialized agents for web scraping, social media intelligence, pricing analysis, and market positioning, delivering automated 360° competitor profiles from multi-source pipelines.
- Engineered full UK GDPR compliance (data minimization, consent management, lawful-basis documentation, right-to-erasure) and owned end-to-end AWS infrastructure (IAM, security, scalable deployment) while leading a cross-functional developer team.

**Kresus — Web3 Conversational AI Trading Agent (Multi-Chain DeFi Copilot)** LangGraph · Claude · FastAPI · SSE Streaming · Solana / Base / Worldchain · Non-Custodial · Tool-Calling · Live: [kresus.com](https://kresus.com)

- Architected the AI orchestration layer for Kresus, a production Web3 conversational trading agent — a FastAPI + LangGraph service driving Claude that lets users research, analyze, and execute on-chain trades across Solana, Base, and Worldchain through natural language, replacing a previous in-process Rust agent.

- Engineered a multi-turn agentic tool-calling loop over 30+ trading tools (token discovery, DexScreener and Hyperliquid market data, candlestick technical analysis, swaps, transfers, portfolio balances, and pump.fun token deployment), streaming responses to the mobile app as Server-Sent Events (SSE).
- Preserved a non-custodial, client-side signing model (tools return prepared, unsigned transactions for the mobile wallet to sign) and enforced deterministic safety guardrails — unsupported-chain and cross-chain-swap guards enforced in code — with structured logging of input, intent, and tool-call signals for full observability.

### **Advanced Hybrid RAG — LightRAG + Neo4j + Azure Document Intelligence** *LightRAG · Neo4j GraphDB · Azure AI · LangGraph · GraphRAG · Live: [auxee.com](https://auxee.com)*

- Built a hybrid multi-agent Retrieval-Augmented Generation (RAG) system combining LightRAG (Neo4j knowledge graph) for entity and relationship extraction with Azure Document Intelligence for structured parsing across all document types.
- Implemented smart routing (LightRAG + SQL for Excel/CSV; Azure Document Intelligence + GraphRAG for PDF/DOCX/TXT), enabling multi-hop reasoning beyond traditional vector search.
- Deployed internally at [Aramco \(Saudi Aramco\)](#) for petroleum-management teams as an enterprise knowledge assistant, enabling staff to query large volumes of technical and operational documents in natural language.

### **NeuroSync — Real-Time AI Meeting Intelligence & Autonomous CRM** *OpenAI Whisper · Speaker Diarization · LangGraph · Salesforce API · AWS Lambda*

- Designed a real-time meeting intelligence platform that transcribes live meetings via OpenAI Whisper with speaker diarization, extracts action items, deal signals, and sentiment, and syncs to Salesforce CRM in under 3 seconds post-meeting.
- Deployed a fully serverless, event-driven architecture on AWS Lambda + API Gateway with sub-200ms processing latency per transcript chunk at concurrent scale.

### **SynthAgent — Autonomous AI Research Analyst (Self-Correcting ReAct)** *ReAct · Self-Reflection · Vertex AI Gemini · LangGraph · Firecrawl*

- Built a ReAct-pattern research agent with self-correction loops that autonomously plans research, evaluates output quality, triggers additional tool calls, and refines findings before delivering citation-rich reports via Vertex AI Gemini.
- Reduced senior-analyst research cycles from 8 hours to under 25 minutes through multi-source Firecrawl + RAG synthesis.

### **Senior AI Engineer — Gicoh**

*Lahore, Pakistan · Full-time · On-site*

*Apr 2023 – Mar 2025*

### **Real-Time Live-Call Conversational Chatbot** *RAG · LLaMA 3.2 · FAISS · Google STT/TTS · Python SIP*

- Architected an end-to-end real-time conversational AI system with a custom RAG pipeline (LLaMA 3.2 + FAISS) for sub-second knowledge retrieval over live inbound phone calls.
- Built a custom Python SIP client integrating the chatbot with an Elastix VoIP PBX, enabling autonomous 24/7 inbound call handling without human intervention.

### **HighTribe — Multi-Agent Travel Assistant (Live on App Store & Google Play)** *LangGraph · Multi-Agent · Gemini · GPT-4 · Google Maps API · Live: [hightribe.com](https://hightribe.com)*

- Designed a production-grade multi-agent system (LangGraph) live on web, Google Play, and Apple App Store, with specialized agents for internal DB querying, Google Maps, and real-time web search.
- Integrated dual-LLM routing (Gemini + GPT-4) with fallback strategies for optimal response quality and cost efficiency at scale.

### **IntelliFlow — Enterprise Autonomous Workflow System** *n8n · LLM Agents · GPT-4 · LLaMA · Workflow Orchestration*

- Architected an enterprise-grade workflow automation platform on n8n integrated with LLM-powered agents (GPT-4, LLaMA) for complex multi-step business process automation without human intervention.
- Designed modular LLM agent pipelines with decision-making, conditional branching, and dynamic task execution across enterprise systems.

### **CareVision — HIPAA-Compliant Healthcare AI Management System** *Healthcare AI · HIPAA · Voice AI · NLP · PHI Security · Live: [carevisioncms.co.uk](https://carevisioncms.co.uk)*

- Developed a HIPAA-compliant healthcare platform with PHI encryption at rest and in transit, RBAC, and audit logging meeting US federal healthcare data-privacy requirements.
- Built an AI care-plan generation engine using LLMs and voice-based data entry (STT/TTS) with NLP medical entity extraction from clinical notes and diagnosis codes.

**VisionGuard — Multimodal AI Fraud Detection (FinTech)** *Multimodal AI · Anomaly Detection · AWS SageMaker · Computer Vision*

- Built a fraud-detection system fusing transaction data, document image analysis (ID/cheque verification), and behavioral biometrics into a unified risk score via a multimodal transformer architecture.
- Deployed on AWS SageMaker auto-scaling endpoints handling 10,000+ transactions/min at p99 latency under 80ms, meeting real-time fraud-prevention SLAs.

**DocuMind — Autonomous AI Legal Document Review Agent** *LegalTech · Azure OpenAI · LangGraph · RAG · NER*

- Architected an autonomous legal-review agent that ingests contracts and NDAs to flag risky clauses, missing obligations, and jurisdiction-specific issues via a LangGraph multi-step reasoning agent (Azure OpenAI GPT-4o) over a legal RAG knowledge base.
- Delivered Critical/High/Medium/Low risk-severity reports, reducing law-firm document review from days to under 10 minutes per document.

**Employee Counting & Tracking — Live CCTV Computer Vision** *YOLOv8 · DeepSORT · FastAPI · RTSP · Face Recognition*

- Built a real-time CV system for live employee counting and tracking from CCTV feeds (FastAPI + RTSP) using YOLOv8 + DeepSORT for persistent identity tracking, plus a face-recognition biometric layer for named-presence detection in crowded environments.

**CRM/ERP Data Extraction Agent** *LangGraph · Natural Language to SQL · Custom SQL Tool*

- Building a LangGraph AI agent that lets non-technical users query production ERP databases in natural language (NL-to-SQL) via a custom tool integrated into the agent graph.

**AI Engineer** — Techling (Private) Limited

*Jun 2021 – Apr 2023*

*Lahore, Pakistan · Full-time · On-site*

**Chex.AI — End-to-End Vehicle Inspection AI Platform** *YOLOv8 · AWS · FastAPI · Pixtral LLM · Live: [chex.ai](https://chex.ai)*

- Delivered a full-stack AI vehicle-inspection platform (FastAPI backend, AWS authentication + S3 storage) with license-plate extraction and vehicle-damage detection using fine-tuned YOLOv8.
- Integrated the Pixtral multimodal LLM for human-readable damage-assessment reports, plus a video object-tracking module for detecting damage across live video streams.

**DeftGPT — Multi-LLM Unified AI Platform (11 LLMs, AWS)** *FastAPI · AWS · Multi-LLM Orchestration · GPT-4 · LLaMA · Live: [deftgpt.com](https://deftgpt.com)*

- Built and deployed a unified platform integrating 11 LLMs through a single FastAPI backend, allowing users to compare, switch, and receive parallel responses from multiple models on AWS.
- Architected full LLM API routing, user authentication, and session management for advanced multi-model orchestration.

**Trainee Data Scientist** — Programmers Force

*Mar 2021 – Jun 2021*

*Lahore, Pakistan · Full-time · On-site*

- Contributed to hands-on machine learning model development and applications under experienced mentorship in a cross-functional team environment.

## EDUCATION

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**MS Data Science (AI/ML)** — AUIC Islamabad

*CGPA 3.12/4 · Feb 2020 – Jan 2023*

**BS Computer Science** — GC University Faisalabad

*CGPA 3.42/4 · Sep 2015 – Aug 2019*

## CERTIFICATIONS

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Machine Learning Specialization — Coursera (Stanford University / DeepLearning.AI), Andrew Ng